Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS

[CGPA-3.32] in Cycle 1

Academic Year 2026-27
FY MCA- SEM I-DIV A
IT12 Data Structure and Algorithms
Practical Assignment1

Roll No : 2601008 **Name : Prasad Devidas Bairagi**
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Create a Singly Linear Linked List with following operations:

1. **Create Linked List-append (create nodes with positive and negative numbers)**
2. **Traverse and print the node values**
3. **Insert node at a specific position**
4. **Delete the nodes having negative values**

```
class Node:
    def __init__(self, val):
        self.data = val
        self.next = None

class LinkedList:
    def __init__(self):
        self.head = None

    # 1. Create Linked List - append
    def append(self, new_node):
        if self.head is None:
            self.head = new_node
        else:
            temp = self.head
            while temp.next:
                temp = temp.next
            temp.next = new_node

    # 2. Traverse and print the node values
    def print(self):
        temp = self.head
        if not temp:
            print("Linked List is empty")
            return
        while temp:
            print(temp.data, end=" -> ")
```

```
            temp = temp.next
        print("None")

    # 3. Insert node at a specific position (1-based index)
    def insert(self, new_node, position):
        if position < 1:
            print("Invalid position")
            return
        if position == 1:
            new_node.next = self.head
            self.head = new_node
            return
        temp = self.head
        count = 1
        while temp and count < position - 1:
            temp = temp.next
            count += 1
        if not temp:
            print("Position out of range")
            return
        new_node.next = temp.next
        temp.next = new_node

    # 4. Delete the nodes having negative values
    def delete_negative(self):
        while self.head and self.head.data < 0:
            self.head = self.head.next
        temp = self.head
        while temp and temp.next:
            if temp.next.data < 0:
                temp.next = temp.next.next
            else:
                temp = temp.next

list = LinkedList()

list.append(Node(10))
list.append(Node(-5))
list.append(Node(20))
list.append(Node(-15))
```

```
list.append(Node(30))

print("Original Linked List:")
list.print()

print("\nInsert node at a specific position :")
list.insert(Node(99), 3)
list.print()

print("\nAfter deleting negative values:")
list.delete_negative()
list.print()
```

OUTPUT :

```
PS D:\> cd '.\IMCC ADM\Assignment\DSA'

PS D:\> cd '.\IMCC FYMCA\Assignment\DSA\'
PS D:\IMCC FYMCA\Assignment\DSA> python SinglyLinearLinkedList.py
Original Linked List:
10 -> -5 -> 20 -> -15 -> 30 -> None

Insert node at a specific position :
10 -> -5 -> 99 -> 20 -> -15 -> 30 -> None

After deleting negative values:
10 -> 99 -> 20 -> 30 -> None
```